

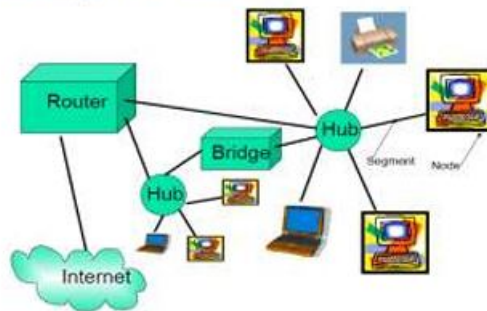
# Chapter 1: Introduction to Computer Network

## Introduction to Computer Network

### What is computer Network?

Computer Network is a collection of computing devices. Devices are connected in various ways in order to communicate and share resources. Usually, the connections between computers in a network are made using physical wires or cables. Sometime connections are wireless, using radio waves or infrared signals. Example, a computer network can consists of a collection of computers, printers and other equipment like router, switch and servers that is connected together so that they can communicate with each other.

An example of a network



### Uses/Application of Computer Network

Computer network has a broad range of application. It can be summarized in following points:

1. Business Application
2. Home Application
3. Mobile Users
4. Social Issues

### Business Application

#### a. Resourcesharing

The goal is to make all programs, equipment, and especially data available to anyone on the network without regard to the physical location of the resource or the user. An obvious and wide spread example is having a group of office workers share a common printer.

Telephone calls between employees may be carried by the computer network instead of the phone company. This technology is called IP telephony or Voice over IP (VoIP) when Internet technology is used.

b. Desktop sharing lets remote workers see and interact with a graphical computer screen. This makes it easy for two or more people who work far apart to write a report together.

Many companies are doing business electronically, especially with customers and suppliers. This new model is called e-commerce (electronic commerce) and it has grown rapidly in recent years. Airlines, bookstores, and other retailers have discovered that many customers like the convenience of shopping from home.

## Home Application

The biggest reason to buy a home computer was probably for Internet access. Internet access provides home users with connectivity to remote computers. As with companies, home users can access information, communicate with other people, and buy products and services with e-commerce. Access to remote information comes in many forms. It can be surfing the World Wide Web for information or just for fun. Information available includes the arts, business, cooking, government, health, history, hobbies, re-creation, science, sports, travel, and many others.

## Mobile Users

Mobile computers, such as laptop and handheld computers are one of the fastest-growing segments of the computer industry. People on the go often want to use their mobile devices to read and send email, tweet, watch movies, download music, play games, or simply to surf the Web for information. They want to do all of the things they do at home and in the office. Naturally, they want to do them from anywhere on land, sea or in the air.

Wireless networks are of great value to fleets of trucks, taxis, delivery vehicles, and repairpersons for keeping in contact with their home base. Perhaps the key driver of mobile, wireless applications is the mobile phone.

Text messaging or texting is tremendously popular. It lets a mobile phone user type a short message that is then delivered by the cellular network to another mobile subscriber. Since mobile phones know their locations, often because they are equipped with GPS (Global Positioning System) receivers, some services are intentionally location dependent. Mobile maps and directions are an obvious candidate as your GPS-enabled phone and car probably have a better idea of where you are than you do.

## Social Issues

Social networks, message boards, content sharing sites, and a host of other applications allow people to share their views with like-minded individuals. As long as the subjects are restricted to technical topics or hobbies like gardening, not too many problems will arise. The trouble comes with topics that people actually care about, like politics, religion, or sex. Views that are publicly posted may be deeply offensive to some people. Furthermore, opinions need not be limited to text; high-resolution color photographs and video clips are easily shared over computer networks.

Computer networks make it very easy to communicate. They also make it easy for the people who run the network to snoop on the traffic.

There are multi-person messaging services too, such as the Twitter service that lets people send short text messages called “tweets” to their circle of friends or other willing audiences. The Internet can be used by applications to carry audio (e.g. Internet radio stations) and video (e.g. YouTube). One of the most popular social networking sites is Facebook. It lets people update their personal profiles and shares the updates with other people who they have declared to be their friends.

## Advantages and Disadvantages of Computer Network

## Advantages

- File Sharing
- Resource Sharing
  - Device sharing (Printers)
- In-expensive setup
  - Sharing Resource reduce Cost
- Flexible Handling
  - Easy For Network Admin to Manage Resource
  - User can login from any Host and access his/her files
- Increased Resource Capacity
  - By Sharing Resource

## Disadvantages

- Security Concerns
  - Computer within a Network can be Vulnerable
- Malware(Malicious software)
  - Malicious software(worms, viruses, Trojan Horse, Logic bomb, zombie)
- Lack of Robustness
  - Breakdown of Central System disrupt the Entire System
- Needs An Efficient Handler
  - Required Technically qualified Candidate to maintain the system

## Overview of Network Types

- **LAN(Local Area Network)**

A LAN is a network that is used for communicating among computer devices, usually within an office building or home. LAN's enable the sharing of resources such as files or hardware devices that may be needed by multiple users

- **PAN(Personal Area Network)**

PAN network is a computer network that allows devices to communicate around a single person. The network is structured for a single entity which can be in a small office, a building or apartment. PAN network connects different peripherals such as computers, telephones, video game consoles.

- **CAN(Campus Area Network)**

It is a type of network built upon connection of various LAN networks within restricted geographical area. It is also known as **corporate** network. These are interconnected with high speed Ethernet using optical fiber. The range of CAN varies from 1 km to 5 km.

- **MAN(Metropolitan Area Network)**

It is a large computer network that usually spans a city or a large campus. A MAN is optimized for a larger geographical area than a LAN, ranging from several blocks of buildings to entire cities. A MAN might be owned and operated by a single organization, but it usually will be used by many individuals and organizations. Examples of MAN: Telephone company network that provides a high speed DSL to customers and cable TV network.

- **WAN(Wide Area Network)**

WAN covers a large geographic area such as country, continent or even whole of the world. A WAN is two or more LANs connected together. Multiple LANs can be connected together using devices such as bridges, routers, or gateways, which enable them to share data. The world's most popular WAN is the Internet.

- **GAN (Global Area Network)**

A **global network**, such as the internet, is referred to as the Global Area Network (GAN). The internet is, however, not the only computer network of its kind. Internationally operating companies also support local networks that comprise of several WANs and connect company computers across the world. GANs use the fiber optic infrastructure from wide area networks and combine these with **international undersea cables** or **satellite transmissions**.

## Overview of Network Topologies

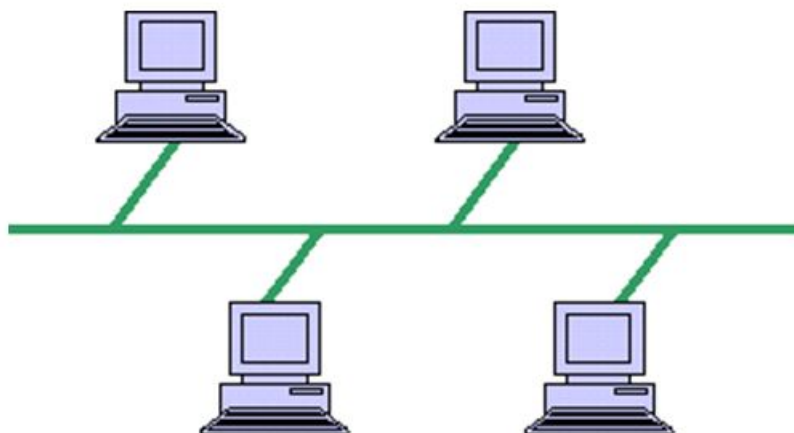
Topology-Physical and logical network layout

- Physical–actually out of the computer cables and other network devices
- Logical– the way in which the network appears to the devices that use it

### Bus Topology

Bus consists of a single linear cable called a trunk. Data is sent to all computers on the trunk.

Each computer examines EVERY packet on the wire to determine who the packet is for and accepts only messages addressed to them.



| Advantages                  | Disadvantages                                       |
|-----------------------------|-----------------------------------------------------|
| Cheap and easy to implement | Network disrupts when computer are added or removed |

|                    |                                                   |
|--------------------|---------------------------------------------------|
| Require less cable | A break in the cable will affect all the networks |
|--------------------|---------------------------------------------------|

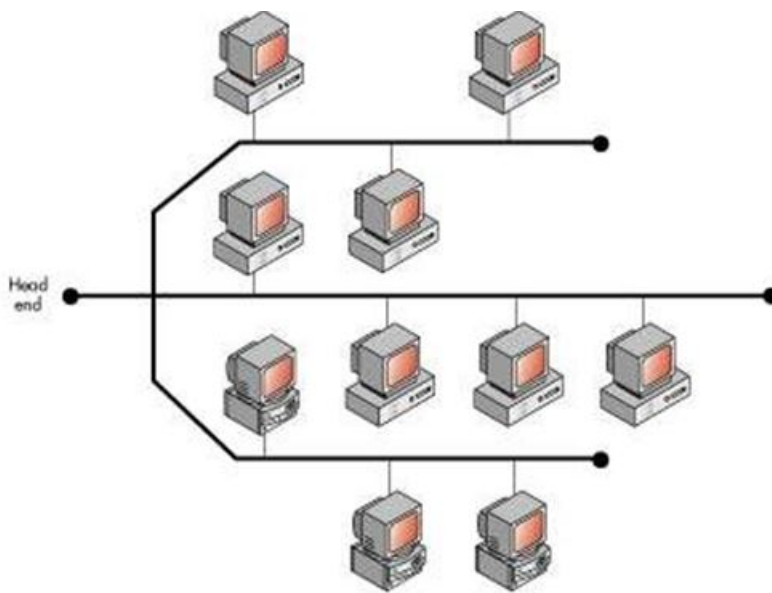
## Distributed Bus

A more complex form of the physical bus topology is the distributed bus.

In the distributed bus, the trunk cable starts at what is called a "root" or "head end", and branches at various points along the way.

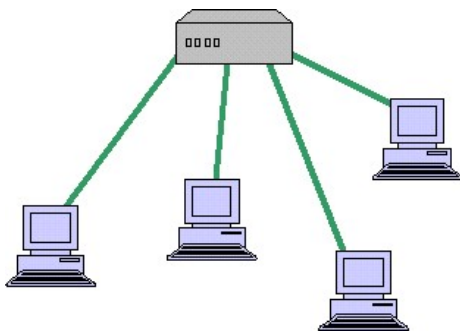
Unlike the simple bus topology, this variation uses a trunk cable with more than two endpoints. Where the trunk cable branches, the division is made by means of a simple connector.

This topology is susceptible to bottle necking and single-point failure.



## Star Topology

- All computers/devices connect to a central device called hub or switch.
- Each device requires a single cable
- Point-to-point connection between the device and hub.
- Most widely implemented

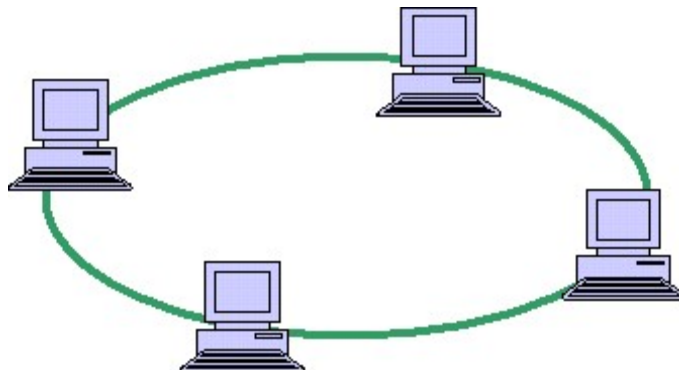


| Advantages                                        | Disadvantages                              |
|---------------------------------------------------|--------------------------------------------|
| Easily expanded without disruption to the network | Requires more cable                        |
| Cable failure affects only a single user          | Centralized device-Single point of failure |

## Ring Topology

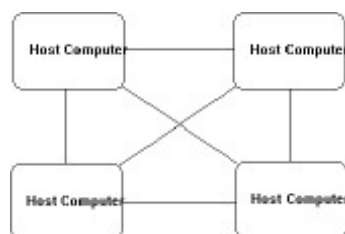
- Computers are connected on a single circle of cable.
- Usually seen in a Token Ring or FDDI (fiber optic) network.

| Advantages                                                                                                        | Disadvantages                                                                                              |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Cable fault are easily located making trouble shooting easier</li> </ul> | <ul style="list-style-type: none"> <li>• Expansion to the network can cause network disruption</li> </ul>  |
| <ul style="list-style-type: none"> <li>• Easy to install</li> </ul>                                               | <ul style="list-style-type: none"> <li>• A single break in the cable can disrupt entire network</li> </ul> |



## Mesh Topology

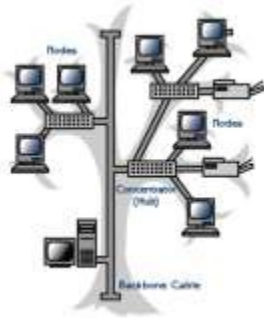
- Each computer connects to every other.
- High level of redundancy.



| Advantages                                 | Disadvantages                                    |
|--------------------------------------------|--------------------------------------------------|
| Provide redundant path between devices     | Require more cable than the other LAN topologies |
| Network can be expanded without disruption | Complicated implementation                       |

## Tree Topology

- A tree topology (hierarchical topology) can be viewed as a collection of star networks arranged in a hierarchy.
- This tree has individual peripheral nodes which are required to transmit to and receive from one other only and are not required to act as repeaters or regenerators.



| Advantages                                                                                          | Disadvantages                                                                                                |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Expansion of network is easy</li></ul>                      | <ul style="list-style-type: none"><li>• Relies on main bus cable, if it breaks entire system fails</li></ul> |
| <ul style="list-style-type: none"><li>• Easy error detection and correction</li></ul>               | <ul style="list-style-type: none"><li>• Slow as number of devices increases</li></ul>                        |
| <ul style="list-style-type: none"><li>• If one segment is affected other are not affected</li></ul> |                                                                                                              |

## Hybrid Topology

- A combination of any two or more network topologies.
- A hybrid topology always accrues when two different basic network topologies are connected.

## CLIENT SERVER MODEL

A client-server network consists of a number of devices called clients who acquire services from the relatively powerful devices called servers. Client devices are typically PCs or mobile devices with network software applications installed. Server device typically stores files, databases and applications like Web sites. The client-server model can be used on the Internet as well as local area networks (LANs). Examples of client-server systems on the Internet include Web browsers and Web servers, FTP clients and servers, and DNS.

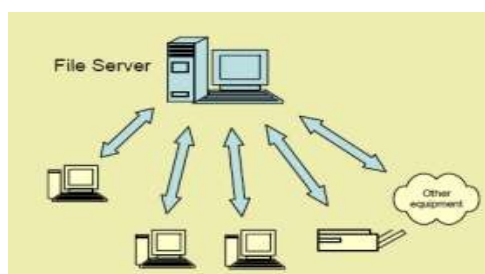


Figure: Client Server model

If we look at the client-server model in detail, we see that two processes (i.e., running programs) are involved, one on the client machine and one on the server machine. Communication takes the form of the client process sending a message over the network to the server process. The client process then waits for a reply message. When the server process gets the request, it performs the requested work or looks up the requested data and sends back a reply. These messages are shown in figure below.

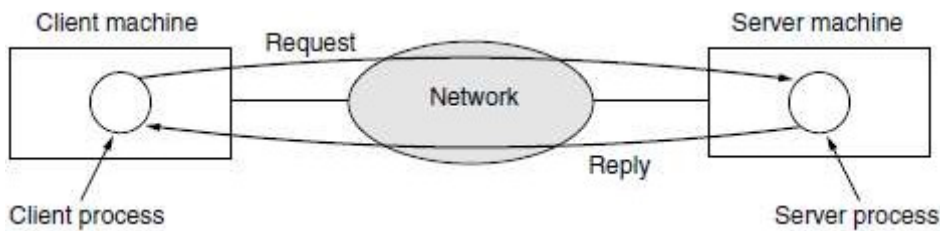


Figure: request and replies in client server model

### Client and its function

**Thin client**– a personal computer that does not have to be very powerful because it only presents the user interface to the user. It is largely used for interaction with processing layers. Example: Increasingly PDA's, Handhelds or Smart phones

**Fat client**–a typically powerful personal computer capable of independent application processes. Also notebook computer or workstation

### Different types of server and function

**Database server**–a server that hosts one or more databases, (Executing all data manipulation commands at the server)

**Transaction server**– a server that hosts services which ensure that all database updates for a transaction succeed or fail as a whole. (Crucial in banking contexts –cash machines; online shopping ...)

**Application server**– a server that hosts application logic and services for an information system. (between interface & database)

**Messaging or groupware server**– a server that hosts services for e-mail, calendaring, and other work group functionality.

**Web server**–a server that hosts Internet or intranet websites, (sends data (XML) and documents (HTML) to clients)



## PEER-TO-PEER NETWORK

P2P is a distributed network architecture, which consists of devices known as peers that share their own resources (disk storage, processing power or network bandwidth) directly, without the need for central server. Peer-to-peer is popular for file sharing on the Internet. Peer-to-peer systems often implement an Application Layer overlay network on top of the native or physical network topology.

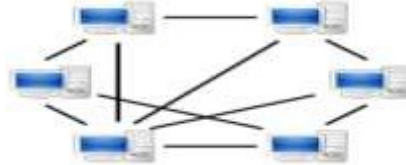


Figure: Peer-to-Peer Network

Peer-to-Peer computing is inspired by the controversial music-sharing service Napster. Instead of Internet information being held in a few central locations, Peer-to-Peer computing makes it theoretically possible to access the files and data residing on every personal computer connected to the Internet.

### Problem with Server-Client Model

- **Scalability**

As the number of user's increases, there is a higher demand for computing power, storage space, and bandwidth associated with the server-side

- **Reliability**

The whole network will depend on the highly loaded server to function properly.

### Advantage of P2P model

- The system is based on the direct communication between peers
- There is zero reliance on centralized service or resources for operations
- The system can survive extreme changes in network composition
- They thrive in a network with heterogeneous environment
- This model is highly scalable

## Peer-to-Peer Networks vs Client/Server Networks

| Peer-to-Peer Networks                                                                                                                                                                        | Client/Server Networks                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Easy to set up</li><li>• Less expensive to install</li></ul>                                                                                         | <ul style="list-style-type: none"><li>• More difficult to set up</li><li>• More expensive to install</li></ul>                                                                                                                 |
| <ul style="list-style-type: none"><li>• Can be implemented on a wide range of operating systems</li></ul>                                                                                    | <ul style="list-style-type: none"><li>• A variety of operating systems can be supported on the client computers, but the server needs to run an operating system that supports networking</li></ul>                            |
| <ul style="list-style-type: none"><li>• More time consuming to maintain the software being used (as computers must be managed individually)</li></ul>                                        | <ul style="list-style-type: none"><li>• Less time consuming to maintain the software being used (as most of the maintenance is managed from the server)</li></ul>                                                              |
| <ul style="list-style-type: none"><li>• Very low levels of security supported or none at all. These can be very cumbersome to set up, depending on the operating system being used</li></ul> | <ul style="list-style-type: none"><li>• High levels of security are supported, all of which are controlled from the server. Such measures prevent the deletion of essential system files or the changing of settings</li></ul> |
| <ul style="list-style-type: none"><li>• Ideal for networks with less than 10 computers</li></ul>                                                                                             | <ul style="list-style-type: none"><li>• No limit to the number of computers that can be supported by the network</li></ul>                                                                                                     |
| <ul style="list-style-type: none"><li>• Does not require a server</li></ul>                                                                                                                  | <ul style="list-style-type: none"><li>• Requires a server running a server operating system</li></ul>                                                                                                                          |
| <ul style="list-style-type: none"><li>• Demands a moderate level of skill to administer the network</li></ul>                                                                                | <ul style="list-style-type: none"><li>• Demands that the network administrator has a high level of IT skills with a good working knowledge of a server operating system</li></ul>                                              |

### ISPs

An ISP, or Internet Service Provider, is an organization that provides businesses and consumers with internet access. They act as a gateway, connecting users to their network and allowing access to the internet. ISPs also provide additional services like email, web hosting, and domain name registration.

#### Key aspects of ISPs:

##### 1. Access Provider:

ISPs provide the infrastructure and network connectivity necessary for users to access the internet.

## **2. Interconnected Networks:**

ISPs are interconnected, allowing users to access different websites and online services across various ISP networks.

## **3. Variety of Services:**

In addition to internet access, ISPs often offer supplementary services such as email, web hosting, and domain name registration.

## **4. Connection Types:**

ISPs offer different types of internet connections, including cable, fiber, DSL, Wi-Fi, and satellite, depending on user needs and geographical locations.

### **Examples in Nepal:**

Nepal Telecom (NTC) and [Websurfer Nepal](#) are examples of ISPs in Nepal

## **NSPs**

In networking, an NSP, or Network Service Provider, is an organization that provides backbone infrastructure and services to facilitate internet connectivity and data transmission. They operate at a higher tier in the network hierarchy than ISPs, offering services like high-capacity data transport and access to the internet. NSPs are crucial for the global internet, ensuring efficient and reliable data flow.

Example of an NSP:

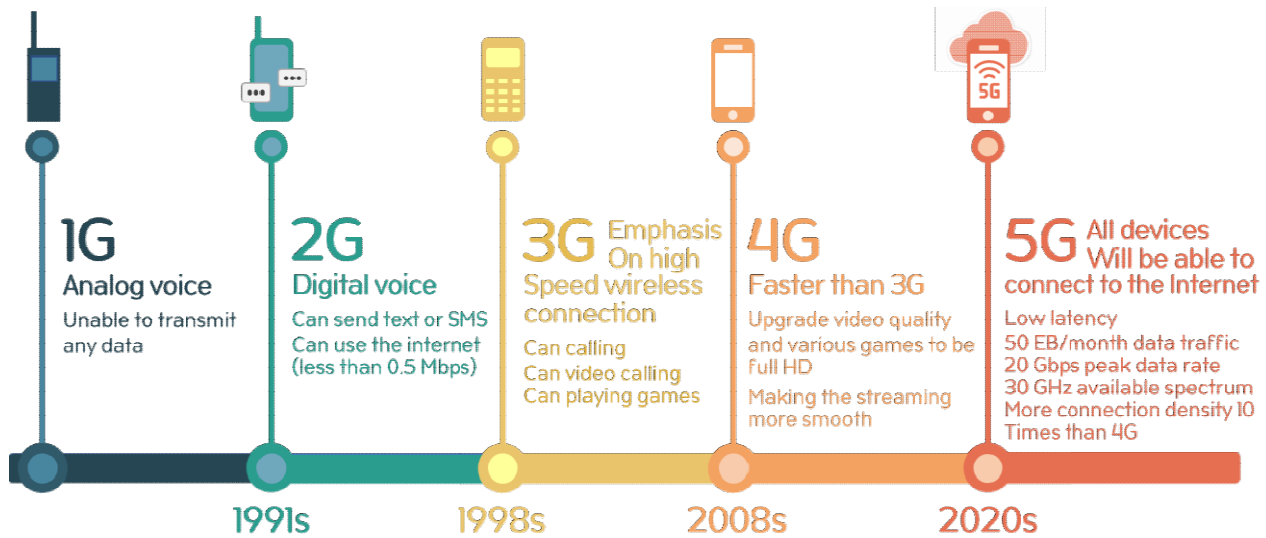
- A large telecommunications company that owns and operates a vast network of fiber optic cables and routers.

### **How NSPs Differ from ISPs:**

- NSPs are typically at a higher tier in the network hierarchy than ISPs.
- ISPs offer internet access directly to consumers and businesses, while NSPs primarily sell their services to other service providers.
- NSPs focus on the backbone infrastructure, while ISPs focus on the "last mile" connectivity to end users.

## Recent Trends in Telecom Technologies:

2G, 3G, 4G, and 5G represent successive generations of mobile network technology, each offering faster data speeds and improved capabilities.



### 1. 2G (Second Generation):

Introduced technologies like GSM (Global System for Mobile Communications), focusing primarily on voice communication and basic data services.

### 2. 3G (Third Generation):

Marked the introduction of mobile internet with technologies like UMTS (Universal Mobile Telecommunications System) and HSPA (High-Speed Packet Access), enabling faster data speeds for features like browsing, email, and basic video streaming.

### 3. 4G (Fourth Generation):

Improved data speeds with technologies like LTE (Long Term Evolution), allowing for higher-quality video streaming, faster downloads, and smoother online gaming.

### 4. 5G (Fifth Generation):

Offers significantly faster speeds, lower latency (delay), and greater network capacity, enabling applications like autonomous vehicles, virtual reality, and high-resolution video conferencing.

